Floating Point

CSE 238/2038/2138: Systems Programming

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Slides adapted from Bryant & O'Hallaron's slides

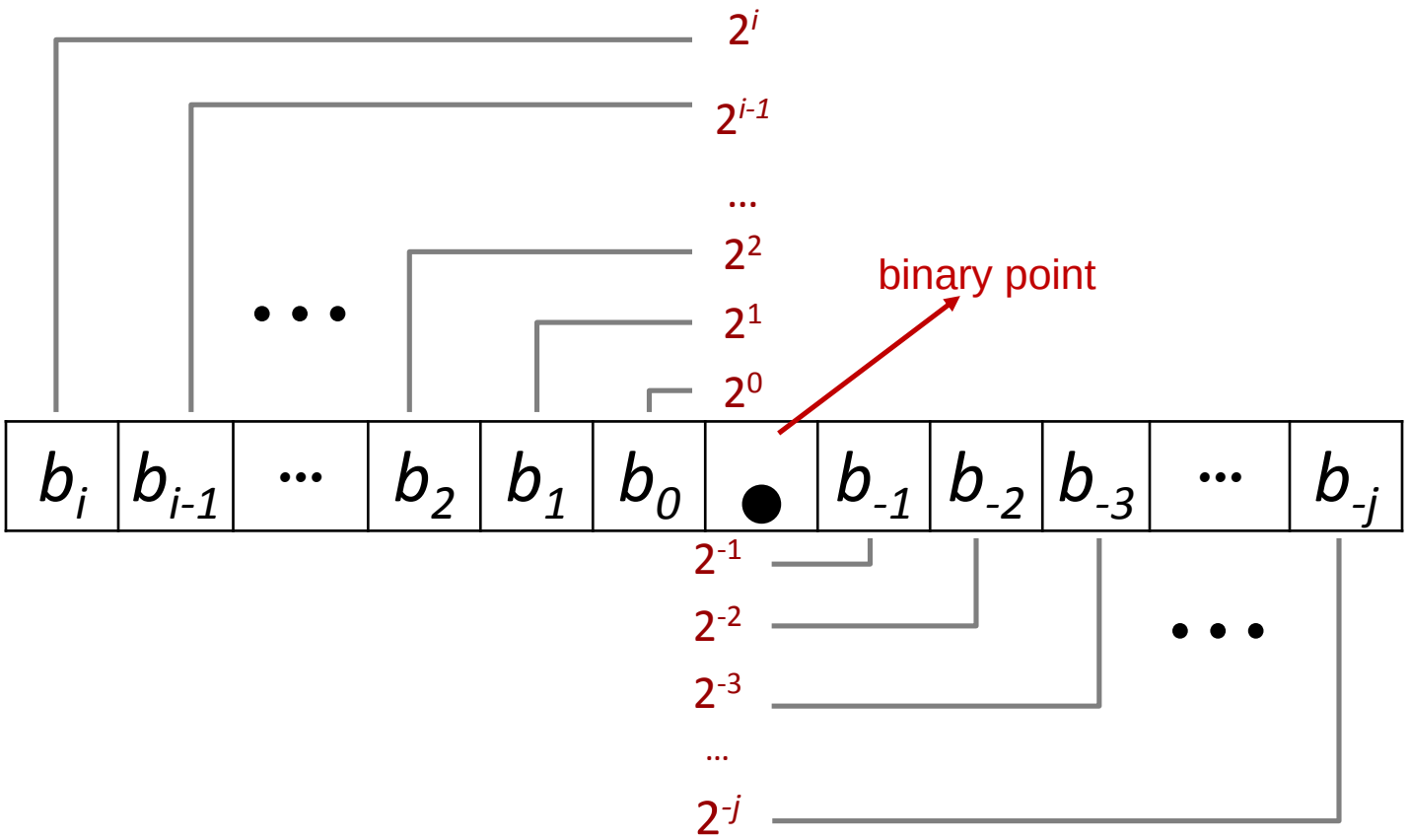
Today: Floating Point

- **Background: Fractional binary numbers**
- IEEE floating point standard: Definition
- Example and properties
- Rounding, addition, multiplication
- Floating point in C
- Summary

Fractional Binary Numbers

What is 1011.101_2?

Fractional Binary Numbers



- **Representation**
  - Bits to right of "binary point" represent fractional powers of 2
  - Represents rational number: $\sum_{k=-j}^{i} b_k \times 2^k$

Fractional Binary Numbers

What is X = 1011.101_2?

$$\sum_{k=-j}^{i} b_k \times 2^k$$

$X = 1*2^3 + 0*2^2 + 1*2^1 + 1*2^0 + 1*2^{-1} + 0*2^{-2} + 1*2^{-3}$

$X = 11\frac{5}{8}$

Fractional Binary Numbers: Examples

Value	Representation
$5\frac{3}{4}$	101.11_2
$2\frac{7}{8}$	010.111_2
$1\frac{7}{16}$	001.0111_2

- **Observations**
  - Divide by 2 by shifting right (unsigned)
  - Multiply by 2 by shifting left
  - Numbers of form $0.111111\ldots_2$ are just below 1.0
    - $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \ldots + \frac{1}{2^i} + \ldots \rightarrow 1.0$
    - Use notation $1.0 - \varepsilon$

Representable Numbers

- **Limitation #1**
  - Only numbers of the form $\frac{x}{2^k}$ can be represented exactly
    - Other rational numbers have repeating bit representations

Value	Representation
$\frac{1}{3}$	$0.0101010101[01]\ldots_2$
$\frac{1}{5}$	$0.001100110011[0011]\ldots_2$
$\frac{1}{10}$	$0.0001100110011[0011]\ldots_2$

- **Limitation #2**
  - Just one setting of binary point within the *w* bits
    - Limited range of numbers (very small values? very large?)

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IEEE Floating Point

- **IEEE Standard 754**
  - Established in 1985 as uniform standard for floating point arithmetic
    - Before that, many machine specific formats
  - Supported by all major CPUs

- **Driven by numerical concerns**
  - Nice standards for rounding, overflow, underflow
  - Hard to make fast in hardware
    - Numerical analysts predominated over hardware designers in defining standard

IEEE Floating Point Representation

Example: (12345_{10} = 0011000000111001_2)
$12345.0_{10} = (-1)^0 * 1.1000000111001_2 * 2^{13}$

- **Numerical Form:**

$$(-1)^s \times M \times 2^E$$

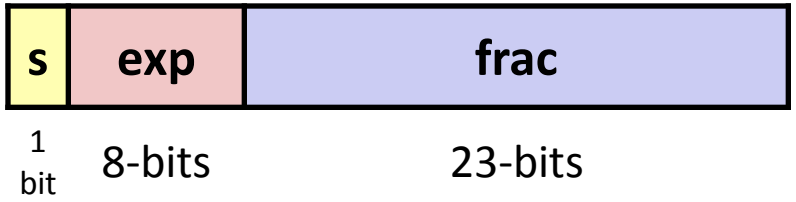
  - **Sign bit** ***s*** determines whether number is negative or positive
  - **Significand** (mantissa) ***M*** normally a fractional value in range
    - [1.0, 2.0), or
    - [0.0, 1.0)
  - **Exponent** ***E*** weights value by power of two

- **Encoding**
  - MSB **s** is sign bit ***s***
  - **exponent** field encodes ***E*** (but is not equal to E)
  - **fraction** field encodes ***M*** (but is not equal to M)

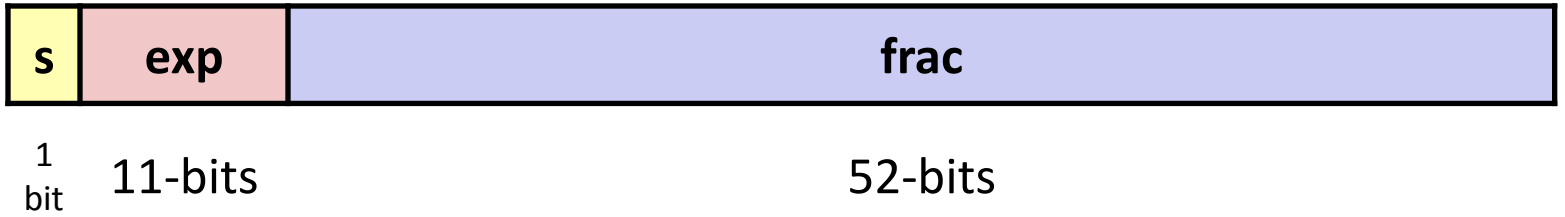


Precision options

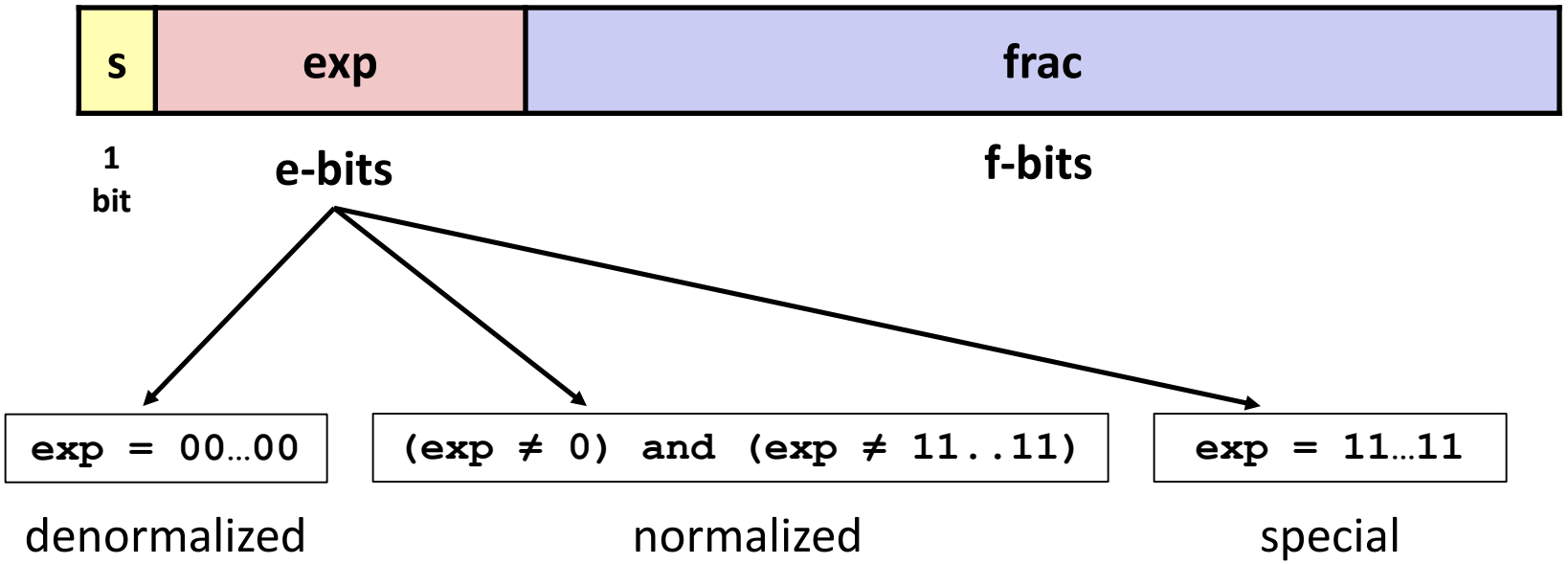
- **Single precision: 32 bits**
  ≈ 7 decimal digits, $10^{\pm 38}$



- **Double precision: 64 bits**
  ≈ 16 decimal digits, $10^{\pm 308}$



Three “kinds” of Floating Point Numbers



"Normalized" Values

$$value = (-1)^s \; M \; 2^E$$
$$E = e - Bias$$

- **When:** `exp ≠ 000…0 and exp ≠ 111…1`

- **Exponent coded as a *biased* value: $E = e - Bias$**
  - *e:* unsigned value of **exp** field
  - $Bias = 2^{k-1} - 1$, where *k* is number of exponent bits
    - **Single precision:** *Bias* = 127 (e: 1…254, E: -126…127)
    - **Double precision:** *Bias* = 1023 (e: 1…2046, E: -1022…1023)

- **Significand (mantissa) coded with implied leading 1: $M = 1.xxx\ldots x_2$**
  - **xxx…x :** bits of *frac* field
  - Minimum when *frac*=000…0 (M = 1.0)
  - Maximum when *frac*=111…1 (M = 2.0 – ε)
  - Get extra leading bit for "free"

Normalized Encoding Example (single prec.)

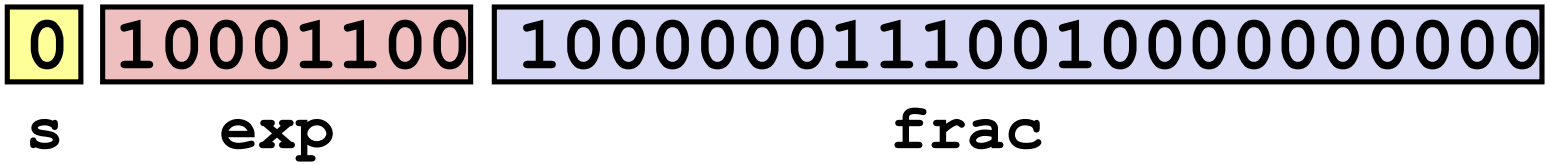
- **Value:** `float F = 12345.0;`
  $12345_{10} = 11000000111001_2$
  $12345.0_{10} = \mathbf{1.1000000111001_2 * 2^{13}}$

value = $(-1)^s$ M 2^E
E = e − Bias

- **Significand**
  M = $1.\underline{1000000111001}_2$
  frac = $\mathbf{10000001110010000000000}_2$ (23 bits)

- **Exponent**
  E = 13
  Bias = 127
  e = 140 = $\mathbf{10001100}_2$ (8 bits)

- **Result:**



“Denormalized” Values

$$value = (-1)^s M 2^E$$
$$E = 1 - Bias$$

- **When: `exp` = `000…0`**
- **Exponent coded as: *E* = 1 – Bias (instead of *E* = e – *Bias*)**
- **Significand coded with implied leading 0: *M* = 0 . `xxx…x`$_2$**
  - **`xxx…x`:** bits of *frac* field
- **Cases**
  - **`exp`** = `000…0`, **`frac`** = `000…0`
    - Represents zero value
    - Note distinct values: +0 and –0 (why?)
  - **`exp`** = `000…0`, **`frac`** ≠ `000…0`
    - Numbers closest to 0.0
    - Equispaced

“Special” Values

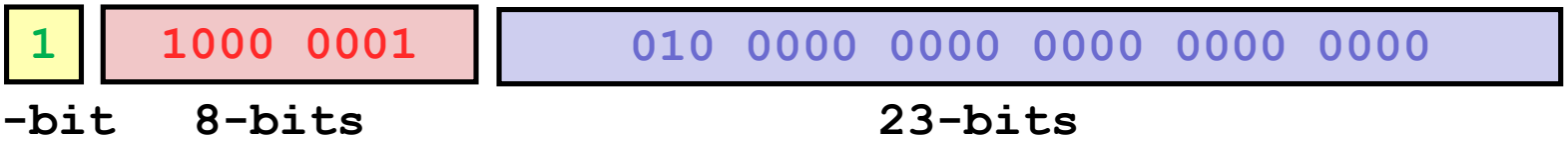
- **When: `exp = 111…1`**
- **Cases**
  - **`exp` = `111…1`, `frac` = `000…0`**
    - Represents value ∞ (infinity)
    - Operation that overflows
    - Both positive and negative
    - e.g., 1.0/0.0 = −1.0/−0.0 = +∞, 1.0/−0.0 = −∞
  - **`exp` = `111…1`, `frac` ≠ `000…0`**
    - Not-a-Number (NaN)
    - Represents case when no numeric value can be determined
    - e.g., sqrt(–1), ∞ − ∞, ∞ × 0

C float Decoding Example

$$value = (-1)^s \; M \; 2^E$$
$$E = e - Bias$$
$$Bias = 2^{k-1} - 1 = 127$$

`float f = 0xC0A00000`

`binary: 1100 0000 1010 0000 0000 0000 0000 0000`



S = 1 → negative number

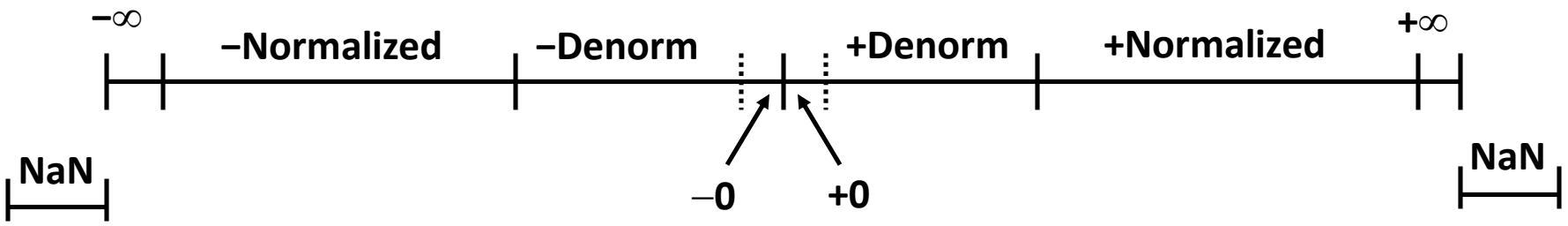
E = 129-127 = 2

M = $1.010\ 0000\ 0000\ 0000\ 0000_2$

= $1 + \frac{1}{4} = 1.25_{10}$

$$value = (-1)^s \; M \; 2^E = (-1)^1 * 1.25 * 2^2 = -5.0$$

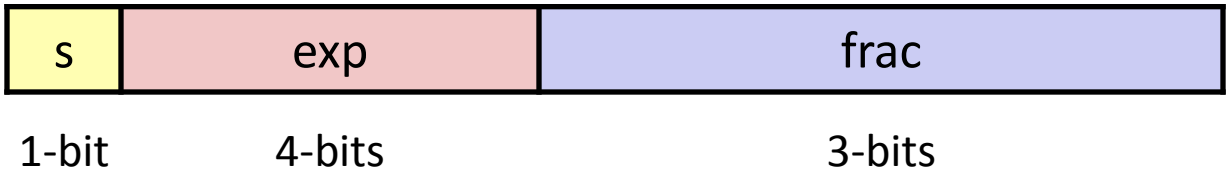
Visualization: Floating Point Encodings



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Tiny Floating Point Example



- **8-bit Floating Point Representation**
  - the sign bit is in the most significant bit
  - the next 4 bits are the **`exp`**, with a bias of 7
  - the last 3 bits are the **`frac`**

- **Same general form as IEEE Format**
  - normalized, denormalized
  - representation of 0, NaN, infinity

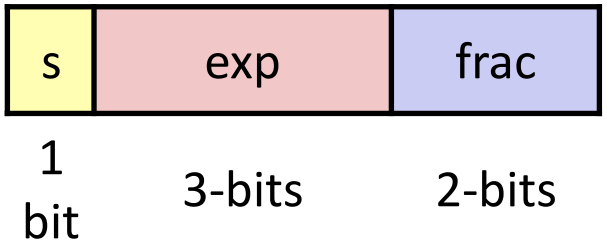
Dynamic Range (8-bit FP numbers)

value = $(-1)^s$ M 2^E
Norm: E = e − Bias
Denorm: E = 1 − Bias

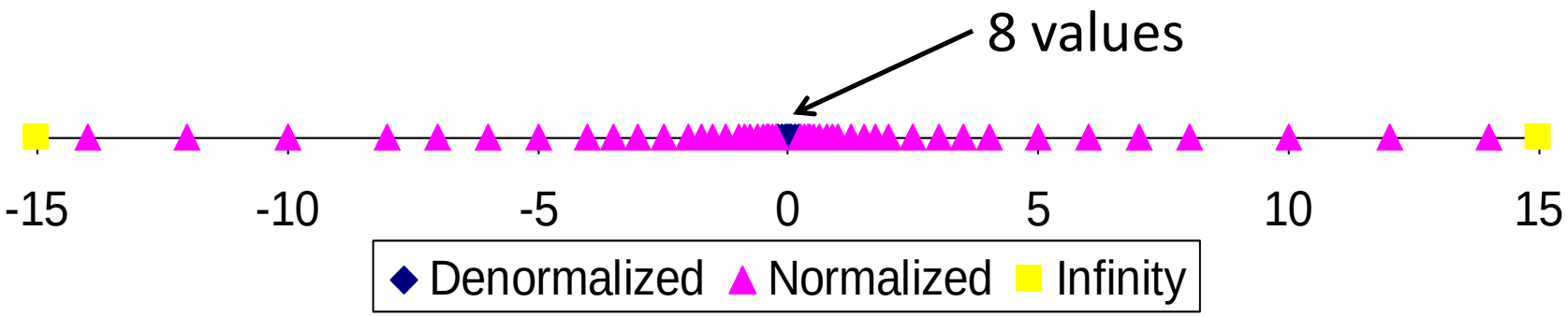
	Bias=7	S	exp	frac	E	Value	Explanation
Denormalized Numbers		0	0000	000	-6	0	
		0	0000	001	-6	1/8 * 1/64 = **1/512**	Closest to zero
		0	0000	010	-6	2/8 * 1/64 = **2/512**	$(-1)^0 * (0+\frac{1}{4}) * 2^{-6}$
						.....	
		0	0000	110	-6	6/8 * 1/64 = **6/512**	$(-1)^0 * (0+\frac{3}{4}) * 2^{-6}$
		0	0000	111	-6	7/8 * 1/64 = **7/512**	Largest denormalized
Normalized Numbers		0	0001	000	-6	1 * 1/64 = **8/512**	Smallest normalized
		0	0001	001	-6	9/8 * 1/64 = **9/512**	$(-1)^0 * (1+\frac{1}{8}) * 2^{-6}$
						.....	
		0	0110	110	-1	14/8 * 1/2 = **14/16**	$(-1)^0 * (1+\frac{3}{4}) * 2^{-1}$
		0	0110	111	-1	15/8 * 1/2 = **15/16**	Closest to one below
		0	0111	000	0	1 * 1 = **1**	$(-1)^0 * (1+0) * 2^0$
		0	0111	001	0	9/8 * 1 = **9/8**	Closest to one above
		0	0111	010	0	10/8 * 1 = **10/8**	$(-1)^0 * (1+\frac{1}{4}) * 2^0$
						.....	
		0	1110	110	7	14/8 * 128 = **224**	$(-1)^0 * (1+\frac{3}{4}) * 2^7$
		0	1110	111	7	15/8 * 128 = **240**	Largest normalized
		0	1111	000	n/a	**inf**	

Distribution of Values

- **6-bit IEEE-like format**
  - e = 3 exponent bits
  - f = 2 fraction bits
  - Bias is $2^{3-1}-1 = 3$

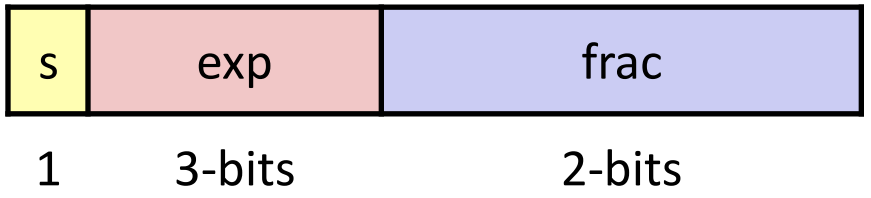


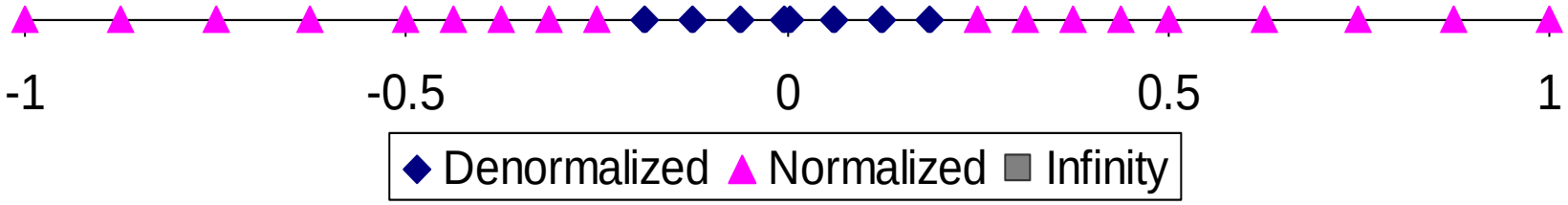
- **Notice how the distribution gets denser toward zero.**



Distribution of Values (close-up view)

- **6-bit IEEE-like format**
  - e = 3 exponent bits
  - f = 2 fraction bits
  - Bias is 3





Special Properties of the IEEE Encoding

- **FP Zero Same as Integer Zero**
  - All bits = 0

- **Can (Almost) Use Unsigned Integer Comparison**
  - Must first compare sign bits
  - Must consider −0 = 0
  - NaNs problematic
    - Will be greater than any other values
    - What should comparison yield?
  - Otherwise OK
    - Denorm vs. normalized
    - Normalized vs. infinity

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Floating Point Operations: Basic Idea

- $x +^f y = \text{Round}(x + y)$

- $x *^f y = \text{Round}(x * y)$

- **Basic idea**
  - First compute exact result
  - Make it fit into desired precision
    - Possibly overflow if exponent too large
    - Possibly round to fit into `frac`

Rounding Modes

VALUE	Towards Zero	Round down (-∞)	Round up (+∞)	Nearest Even (default)
1.40	1	1	2	1
1.60	1	1	2	2
1.50	1	1	2	2
2.50	2	2	3	2
-1.5	-1	-2	-1	-2

Closer Look at Round-To-Nearest-Even

- **Default Rounding Mode**
  - Hard to get any other kind without dropping into assembly
  - All others are statistically biased
    - Sum of set of positive numbers will consistently be over- or under-estimated
- **Applying to Other Decimal Places / Bit Positions**
  - When exactly halfway between two possible values
    - Round so that least significant digit is even
  - Examples: round to nearest hundredth (2 digits after the decimal point)

Value	Action	Rounded Value
7.8949999	Less than half way	7.89
7.8950001	Greater than half way	7.90
7.8950000	Half way—round up	7.90
7.8850000	Half way—round down	7.88

Rounding Binary Numbers

- **Binary Fractional Numbers**
  - "Even" when least significant bit is 0
  - "Half way" when bits to right of rounding position = $100\ldots_2$

- **Examples**
  - Round to nearest 1/4 (2 bits after the binary point)

Value	Binary	Rounded	Action	Rounded Value
$2\frac{3}{32}$	10.00011_2	10.00_2	$<\frac{1}{2}$ → down	2
$2\frac{3}{16}$	10.00110_2	10.01_2	$>\frac{1}{2}$ → up	$2\frac{1}{4}$
$2\frac{7}{8}$	10.111_2	11.00_2	$=\frac{1}{2}$ → up	3
$2\frac{5}{8}$	10.101_2	10.10_2	$=\frac{1}{2}$ → down	$2\frac{1}{2}$

Floating Point Addition

$$(-1)^{s} M \ 2^{E} = (-1)^{s1} M1 \ 2^{E1} + (-1)^{s2} M2 \ 2^{E2}$$

1. **Compare exponents (*E1* and *E2*)**
   - shift the binary point of the number with smaller exponent to the left until the exponents are equal.
2. **Add the significands (*M1* and *M2*) with binary points aligned**
3. **Fix the result**
   a) If $M \geq 2$, shift M right and increment E
   b) If $M < 1$, shift M left k positions, decrement E by k
   c) Overflow if E out of range
   d) Round M to fit `frac` precision

Example: $1.010_2 * 2^2 + 1.110_2 * 2^3 = (0.1010_2 + 1.1100_2) * 2^3$
$= 10.0110_2 * 2^3 = 1.00110_2 * 2^4 = 1.010_2 * 2^4$

E1(2) < E2(3) → 0.1010 + 1.1100 = 10.0110

1 | 2 | 3.a | 3.d

M(10.0110) ≥ 2 → M=round(1.00110) → M=1.010, E=4

Mathematical Properties of FP Addition

- **Closed under addition?** ***Yes***
  - But may generate infinity or NaN
- **Commutative?** ***Yes***
  - `3.14 + 1e10 = 1e10 + 3.14`
- **Associative?** ***No***
  - Overflow and inexactness of rounding
  - `(3.14+1e10)-1e10 = 0, 3.14+(1e10-1e10) = 3.14`
- **0 is additive identity?** ***Yes***
- **Every element has additive inverse?** ***Almost***
  - Yes, except for infinities & NaNs
- **Monotonicity**
  - $a \geq b \Rightarrow a+c \geq b+c$? ***Almost***
    - Except for infinities & NaNs

Floating Point Multiplication

$$(-1)^s M \; 2^E = (-1)^{s1} M1 \; 2^{E1} * \; (-1)^{s2} M2 \; 2^{E2}$$

1. **Exact Result**
   - Sign *s*: *s1* ^ *s2*
   - Significand *M*: *M1* * *M2*
   - Exponent *E*: *E1* + *E2*

2. **Fix the result**
   - If *M* ≥ 2, shift *M* right and increment *E*
   - Overflow if *E* out of range
   - Round *M* to fit **`frac`** precision

Example: $1.010_2 * 2^2 * 1.110_2 * 2^3 = 10.001100_2 * 2^5$

$= 1.0001100_2 * 2^6 = 1.001_2 * 2^6$

Mathematical Properties of FP Mult

- **Closed under multiplication?** ***Yes***
  - But may generate infinity or NaN
- **Commutative?** ***Yes***
- **Associative?** ***No***
  - Possibility of overflow, inexactness of rounding
  - `(1e20*1e20)*1e-20=inf`, `1e20*(1e20*1e-20)=1e20`
- **1 is multiplicative identity?** ***Yes***
- **Multiplication distributes over addition?** ***No***
  - Possibility of overflow, inexactness of rounding
  - `1e20*(1e20-1e20)=0.0`, `(1e20*1e20)-(1e20*1e20)=NaN`
- **Monotonicity**
  - $a \geq b$ & $c \geq 0 \Rightarrow a * c \geq b * c$? ***Almost***
    - Except for infinities & NaNs

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Floating Point in C

- **C Guarantees Two Levels**
  - **`float`** single precision
  - **`double`** double precision

- **Conversions/Casting**
  - Casting between **`int`**, **`float`**, and **`double`** changes bit representation
  - **`double`**/**`float`** → **`int`**
    - Truncates fractional part
    - Like rounding toward zero
    - Not defined when out of range or NaN: Generally sets to TMin
  - **`int`**/**`float`** → **`double`**
    - Exact conversion, as long as **`int`** has ≤ 53 bit word size
  - **`int`** → **`float`**
    - Will round according to rounding mode

Floating Point C Puzzles

Initialization

```
int x = …;

float f = …;

double d = …;
```

Assume neither
`d` nor `f` is NaN

`x == (int)(float) x`	

Floating Point C Puzzles

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`f == (float)(double) f`	True
`d == (double)(float) d`	

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`x == (int)(float) x`	False
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`f == -(-f);`	

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`f == -(-f);`	True
`2/3 == 2/3.0`	

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`2/3 == 2/3.0`	False
`d < 0.0 ⇒ ((d*2) < 0.0)`	

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`f == -(-f);`	True
`2/3 == 2/3.0`	False
`d < 0.0 ⇒ ((d*2) < 0.0)`	True
`d > f ⇒ -f > -d`	

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double d = …;
```

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d nor **f** is NaN

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`d == (double)(float) d`	False
`f == -(-f);`	True
`2/3 == 2/3.0`	False
`d < 0.0 ⇒ ((d*2) < 0.0)`	True
`d > f ⇒ -f > -d`	True
`d*d >= 0.0`	

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`d < 0.0 ⇒ ((d*2) < 0.0)`	True
`d > f ⇒ -f > -d`	True
`d*d >= 0.0`	True
`(d+f)-d == f`	

Floating Point C Puzzles

Initialization

```
int x = …;

float f = …;

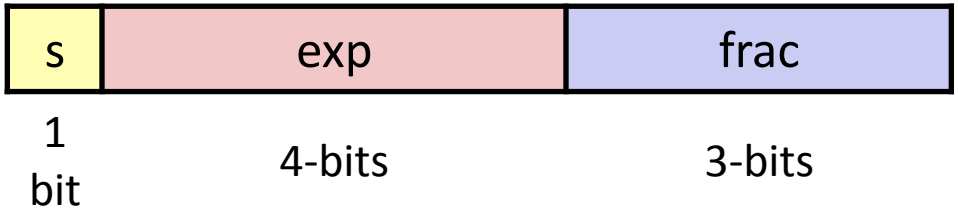
double d = …;
```

Assume neither
d nor **f** is NaN

`x == (int)(float) x`	False
`x == (int)(double) x`	True
`f == (float)(double) f`	True
`d == (double)(float) d`	False
`f == -(-f);`	True
`2/3 == 2/3.0`	False
`d < 0.0 ⇒ ((d*2) < 0.0)`	True
`d > f ⇒ -f > -d`	True
`d*d >= 0.0`	True
`(d+f)-d == f`	False

Creating Floating Point Number

- **Steps**
  1. Normalize to have leading 1
  2. Round to fit within fraction
  3. Postnormalize to deal with effects of rounding

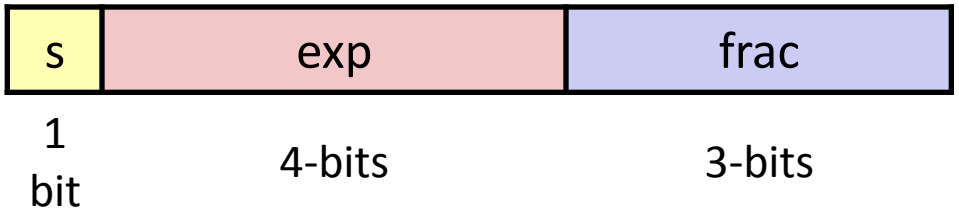


- **Case Study: Convert numbers to tiny floating point format**

Value	Binary
128	10000000_2
15	00001111_2
33	00100001_2
35	00100011_2
63	00111111_2

Step 1: Normalize to have leading 1

- **Requirement**
  - Set binary point so that numbers of form 1.xxxxx
  - Adjust all to have leading one
    - Decrement exponent as shift left

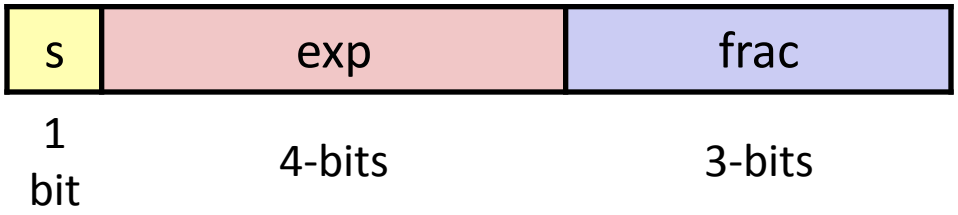


Value	Binary	Fraction	Exponent
128	10000000_2	1.0000000_2	7
15	00001111_2	1.1110000_2	3
33	00100001_2	1.0000100_2	5
35	00100011_2	1.0001100_2	5
63	00111111_2	1.1111100_2	5

Step 2: Round to fit within fraction

Round-to-nearest-even

frac = 3 bits



Value	Fraction	Rounded	Exponent
128	1.0000000_2	1.000_2	7
15	1.1110000_2	1.111_2	3
33	1.0000100_2	1.000_2	5
35	1.0001100_2	1.001_2	5
63	1.1111100_2	10.000_2	5

Step 3: Postnormalize to deal with effects of rounding

- **Issue**
  - Rounding may have caused overflow
  - Handle by shifting right once & incrementing exponent

Value	Rounded	Exponent	Adjusted	Adjusted Exponent	Result
128	1.000_2	7		7	$1.000_2 * 2^7 = 128$
15	1.111_2	3		3	$1.111_2 * 2^3 = \frac{15}{8} * 2^3 = 15$
33	1.000_2	5		5	$1.000_2 * 2^5 = 32$
35	1.001_2	5		5	$1.001_2 * 2^5 = \frac{9}{8} * 2^5 = 36$
63	10.000_2	5	1.000_2	6	$1.000_2 * 2^6 = 64$

Binary Representations

$$value = (-1)^s \; M \; 2^E$$
$$E = e - Bias$$
$$Bias = 2^{k-1} - 1 = 7$$

Unsigned Value	Mantissa	Exponent	Floating Point Value	Binary representation
128	1.000_2	7	128	0 1110 000
15	1.111_2	3	15	0 1010 111
33	1.000_2	5	32	0 1100 000
35	1.001_2	5	36	0 1100 001
63	1.000_2	6	64	0 1101 000

Summary

- **IEEE Floating Point has clear mathematical properties**
- **Represents numbers of form M x 2^E**
- **One can reason about operations independent of implementation**
  - As if computed with perfect precision and then rounded
- **Not the same as real arithmetic**
  - Violates associativity/distributivity
  - Makes life difficult for compilers & serious numerical applications programmers

Interesting Numbers {single,double}

Description	*exp*	*frac*	*Numeric Value*
Zero	**00…00**	**00…00**	**0.0**
Smallest Pos. Denorm.	**00…00**	**00…01**	$2^{-\{23,52\}} \times 2^{-\{126,1022\}}$
▪ Single ≈ 1.4×10^{-45}			
▪ Double ≈ 4.9×10^{-324}			
Largest Denormalized	**00…00**	**11…11**	$(1.0 - \varepsilon) \times 2^{-\{126,1022\}}$
▪ Single ≈ 1.18×10^{-38}			
▪ Double ≈ 2.2×10^{-308}			
Smallest Pos. Normalized	**00…01**	**00…00**	$1.0 \times 2^{-\{126,1022\}}$
▪ Just larger than largest denormalized			
One	**01…11**	**00…00**	**1.0**
Largest Normalized	**11…10**	**11…11**	$(2.0 - \varepsilon) \times 2^{\{127,1023\}}$
▪ Single ≈ 3.4×10^{38}			
▪ Double ≈ 1.8×10^{308}			